

PRATIMAI · AI HANDOUTS

# Build the AI Cycle

*My case file for planning, testing and improving an AI people can trust*

**CLASS 8 · LEVEL 3 — BUILD THE CYCLE**

**17 PAPER MISSIONS · 5 CHAPTERS · NO SCREEN NEEDED**

**PRATIMA AI Curriculum Team**

How do I know? · Who is missing? · What is the trick here?

THIS CASE FILE BELONGS TO

name · class · roll number

CASE OPENED ON ★ CASE CLOSED ON

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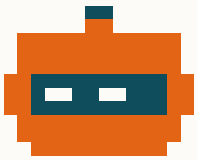
Every activity in this book is **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. All examples use invented people and places.

**THE LICENCE IN PLAIN WORDS**

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Set in **Nunito** and **Baloo 2** on warm paper stock. Both typefaces are used under the SIL Open Font Licence. The book's diagrams are printed as crisp, high-resolution figures, and every page was reviewed individually before this edition went to press.



PRATIMAI · AI HANDOUTS

# Acknowledgments

To the Classes 6 and 7 investigators whose case files came back covered in pencil, arrows and honest confidence circles — this book is built on your evidence. Your teachers told us which missions turned a quiet classroom into a planning meeting, where the instructions needed one more worked example, and which questions made the whole class argue productively for ten minutes. Every note shaped a page here.

This series was built brick by brick by three teams working as one. The **AI research team** checked every explanation against how machines really learn, so nothing in this book is a myth wearing a friendly font. The **academic team** walked every chapter beside the school syllabus, so each game quietly trains a skill that examinations will ask for later. And the **child-psychology team** shaped the pace of every page: short missions for young attention, quick wins that open every chapter, confidence circles that make honesty feel safe, and space to write on every task — because an idea worked by the hand is remembered by the mind. Alongside them, curriculum reviewers guarded the "unplugged" promise: not one mission in this series asks for a phone, an account or a password.

Thanks go as well to the designers of Nunito and Baloo 2, whose friendly letterforms keep pages of hard planning feeling light. And finally, to the AI Engineer holding this book: every blank table, empty checklist and half-drawn cycle map in here was left open on purpose — because on a real project, the plan is never handed to you. The next plan in this file is yours.

17 PAPER MISSIONS · 16 WORDS TO KEEP · 5 CHAPTERS · 1 BADGE EARNED

# How to Use This Handbook

Welcome back, Engineer! Two years of clues sit behind you: in Class 6 you learned to **spot** learning machines, and in Class 7 you operated one — sorting, predicting and watching machines get fooled. This year you take the next step: you **plan and build**. Every serious AI system in the world began as a plan on paper — a sharp problem statement, a list of data, a test, a fair check — exactly the artefacts you will produce in these five chapters. You will still not need a computer, a phone or an app: professional engineers plan on whiteboards and paper before they touch a keyboard, and so will you. Every chapter hands you missions (**tasks**), evidence to examine, and empty spaces to fill with your decisions. This book belongs to you: write, draw, cross out, improve.

**WHY THIS BOOK WORKS THE WAY IT DOES**

This handbook is built around how teenage brains actually learn. Every chapter opens with a quick win — motivation first, theory second. You **do** before you **define**: each idea is met through an experience before it is named, because memory hooks onto action. The confidence circles train metacognition — knowing how sure you are is a skill, and honest rating makes it stronger. Tasks are time-boxed because attention works in sprints; ideas return across chapters on purpose, because a second meeting is what turns exposure into understanding; and every task leaves space to write, because producing your own answer beats re-reading every time. Nothing here can be failed: a wrong answer you go on to check is a clue found early.

**1 READ**

Read the mission twice. Underline anything that puzzles you — puzzles are clues.

**2 DO**

Do the task with your partner, group, class or family, exactly as the labels say.

**3 WRITE**

Write what you decided in the spaces, then shade your confidence circles honestly.

## The work-mode labels

| | |
|---|---|
| **ALONE** | your own case — you investigate by yourself |
| **PAIR** | two engineers — share one book, take turns reading codes aloud |
| **GROUP** | three to four engineers — decide roles before you start |
| **CLASS** | the whole class plays together, led by your teacher |
| **HOME** | a mission for home — show an adult what you discovered |

**HOW TO READ A MISSION**

**T8-01 Mission title** — **PAIR · 15 MIN**

■ **QUICK WIN** marks the easier missions; ■ **HANDS-ON LAB** marks the working labs — missions where you **build, measure, play or draw**.

Every mission has a code like **T8-01** — **T8** means this Class 8 handout, and **01** is the mission number, so your teacher can say “open Task T8-09”. The **PAIR** label tells you who to work with, **15 MIN** is your time limit, and **QUICK WIN** marks an easy first success. This book’s missions ask you to **decide and justify**, not just to sort — so expect to write a reason beside almost every answer. Nobody fails in this book, because every attempt teaches a clue.

**THE QUESTION HABIT — YOUR SUPERPOWER**

Your three question-habit questions — **"How do I know?"**, **"What is missing?"**, **"Who made this?"** — get two powerful companions this year. For every plan you meet (including your own): **"Who is missing?"** — which group was never asked, never counted, never helped? And for every claim a machine or a company makes: **"What is the trick here?"** Engineers who ask these two questions early catch unfair machines **before** people get hurt. Use all five questions in this book, at home, and on every app that promises to be smart.

**DRILL THE QUESTIONS — ON A REAL PLAN**

A vendor pitches your school: "Our AI invigilation camera flags cheating with 98% accuracy." Before anyone signs anything, run the five questions and write one line each:

1. **How do I know?** — Who measured 98%, on which students?

2. **What is missing?** — What does the accuracy number NOT say?

3. **Who made this, who is missing?** — Whose faces was it tested on?

4. **Who answers when it fails?** — Write the sentence you would demand in the contract.

## A Note for Adults

This handout builds **AI understanding** — how machines learn from data and how humans should judge them — through 17 paper-and-pencil missions. It is fully **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. This year the big ideas are the AI project cycle (scope, data, build and test, reflect and improve), data representation and sampling, consent, a nearest-example classifier run by hand, accuracy, no-code tool types and their limits, how text generators predict next words and why they invent, and responsible-use decisions. The best help you can give is to ask the Question-Habit questions, praise careful **reasoning** more than quick right answers, and treat the confidence circles as honest thinking, not scores. A separate teacher pack carries answer notes and rubrics; this book deliberately never prints solutions beside a task — only short hints at the very back.

### Four ways to help — without giving answers

### 1 Ask about the plan

When your engineer is stuck, ask: **"Who exactly has this problem, and what data would show it?"** A sharp question rebuilds a plan faster than a hint.

### 2 Praise the trade-off thinking

Swap "correct!" for **"you noticed the plan helps one group and misses another"** or **"you chose the slower option for a fair reason — that is engineering."**

### 3 Let the plan fail

A plan that fails its test is the most valuable page in the book. Ask **"what will you change in version 2?"** — that loop is the whole subject.

### 4 Connect to real decisions

Point at systems around you — the bus route app, the exam result predictor, the clinic's queue machine — and ask **who is missing** from each one's data.

**IF YOUR CHILD ASKS…**

**"Can I build real AI after this book?"** — Say: you will have built the **plan** the way professionals start — problem, data, test, fairness check. The coding comes years later, and it is the easy-looking part. **"Why does the chatbot make things up?"** — Say: it writes by guessing likely next words, not by checking facts, so a confident lie and a confident truth are built the same way. That is why this book trains verify-first habits. **"Is no-code cheating?"** — Say: no. Choosing the right tool, the right data and the right test **is** the thinking; the tool is just the last step.

**SIGNS OF ENGINEER GROWTH WORTH PRAISING**

Watch for these quiet changes — they matter more than any correct answer: your child **states the problem precisely** before jumping to solutions · they ask **whose data** is in a system and **whose is missing** · they test a plan before trusting it — and improve it after · they say **it depends** and then explain what it depends on · they name a trade-off — speed against fairness, cheap against accurate — and defend a side. When you spot one, name it: "That is exactly what engineers do."

## My AI Journey Map

THE FIVE STRANDS YOU WILL MEET AT EVERY STOP

**U Understand AI**
What is the difference between AI, machine learning and a chatbot?

**D Data**
Who is inside my dataset — and who is missing from it?

**L Learn**
How does a nearest-example learner work — and how do we measure it?

**W World**
Which no-code tool fits which problem — and what are its limits?

**R Responsibility**
Who decides what a machine may do — and who answers for it?

**THINKING-TOOLS THREAD**

runs through every stop: **decompose** the problem · **test before trusting** · **improve in loops** · **optimise under limits**

THE FIVE STOPS ON YOUR CASE

1 **The AI Project Cycle** — *scope · data · build · improve*

2 **Data & Fairness** — *sampling · consent · missing groups*

3 **How Models Learn** — *nearest neighbour · accuracy · tools*

4 **Ethics & Responsible AI** — *next-word guessing · misuse · court*

5 **AI Project Proposal** — *plan it · peer review · optimise*

## BY THE END OF THIS BOOK, I CAN…

- [ ] define **AI** and **machine learning** in my own words and place chatbots, image classifiers and recommenders on a who-learns-what map
- [ ] walk a local problem through the full AI project cycle — on paper
- [ ] hand-run a nearest-example classifier, compute its accuracy on a test set, and suggest one improvement
- [ ] audit a small dataset for missing groups and compute percentages per group
- [ ] explain why collecting data needs **consent** and privacy care
- [ ] match no-code tool types — classifier, chatbot, predictor — to problems, with inputs, outputs and limits
- [ ] explain why a text generator can sound right and still be wrong — and verify its claims
- [ ] apply a fairness checklist and propose fixes for an unfair dataset
- [ ] make and defend a responsible-use decision about privacy, misinformation or social impact
- [ ] break a multi-variable problem into parts and find the best choice under limits

### THREE PLACES I WILL SPOT DATA THIS WEEK

An engineer plans with data that already exists. Name three places data about YOU is produced this week — and tick whether anyone asked:

| | | asked? | aware? |
|---|---|---|---|
| 1. _____ | h(4pt) | Y / N | Y / N |
| 2. _____ | h(4pt) | Y / N | Y / N |
| 3. _____ | h(4pt) | Y / N | Y / N |

DAY 1

## My starting line — a pledge to the plan

2 MIN

Fill this in before Chapter 1. On the last page you will re-read it next to your tallies — the gap between the two is this year's proof.

One real problem near me I would love to scope properly this year:

The data I suspect that problem already produces, without anyone collecting it:

My honesty clause — what I will do when a plan of mine fails its own test:

Signed: _____ h(6pt) Date: _____ h(6pt)

### FIRST IMPRESSIONS — THE ENGINEER'S FLIP TEST

Flip through this book for two minutes, the way an engineer first walks a site. Three things I noticed that I intend to question later:

1. _____ h(4pt)
2. _____ h(4pt)
3. _____ h(4pt)

CARD

## The engineer's five questions — my pocket copy

1. **How do I know?** — who measured, on what data?
2. **What is missing?** — which group never got a say?
3. **Who made this?** — and who profits from my trust?
4. **Who is missing?** — whose faces were never in the training set?
5. **What is the trick here?** — what would the honest test have been?

*The question I will use most this year, and where I first plan to use it:*
_____ *h(4pt)*

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## The boxes in this book

You will meet these boxes on every case. Each one wants something different from you — collect them all.

**MISSION**

A paper mission with a code like **T6-01**. The label tells you who you work with and how many minutes you get. Do it — then write. Writing **is** the mission.

**WORD POWER**

A word worth keeping in your engineer kit. Read it three times today — words are the tools of thinking.

**SELF-CHECK**

Skills to grade yourself on. Shade a confidence circle honestly — honest circles make you learn faster.

**THINK & INK**

A question with lines. There is no single right answer here — your **reasons** are the answer.

Also inside: **MYTH BUSTER** busts a wrong idea many people believe · **ENGINEER'S NOTE** hands you a professional's secret · **HOME LINK** (dashed) is a mission to take home and teach an adult.

# 1. The AI Project Cycle

**GUIDING QUESTION**

*What does it take to build an AI that people can trust?*

You are promoted: from machine-**operator** (Class 7) to machine-**engineer**. But no real AI begins with a machine — it begins with a plan. In this chapter you learn the loop every serious AI team runs: **scope** a sharp problem, **collect** the right data, **build and test** a model, then **reflect and improve**. You will turn vague complaints into precise problem statements, sequence the cycle from cards, and walk a complete Plant Doctor project end to end to find exactly where it went wrong.

**MISSIONS** T8-01 – T8-03
**OUTCOMES** 8.U1 · 8.L1 **STRANDS** L T

*Links: Maths — percentages & data · English — precise definitions · Science — plant health case*

**In this chapter**


1.1 From user to engineer: who learns what ..... 11
1.2 The loop every AI project runs ..... 14
1.3 The Plant Doctor case: a complete walk-through ..... 17


**TERMS YOU'LL BUILD WITH**

machine learning · AI project cycle · problem statement · dataset

**WARM-UP · BEFORE YOU READ** ALONE · 2 MIN

Your school wants "an AI that helps students eat better". Write your honest first reaction: what would you need to know BEFORE anyone builds it? Keep your answer — you will rewrite it after Task T8-01.

**ENGINEER'S TOOLKIT — YOU WILL NEED:** pencil · scissors & envelope for cards · sticky notes · your Class 7 notebook

## 1.1 From user to engineer: who learns what

Last year you trained machines with your own hands, so you already know the secret: a **learning** machine is not given rules — it is given **examples**. This year we name the big idea properly. **Artificial intelligence** is the wide field: any machine that does things we used to need human judgement for. **Machine learning** is the engine inside most modern AI: the machine improves at a task by finding patterns in **examples** (a **dataset**) instead of following rules a person typed. And every AI product you meet is really a machine that learned **one specific thing** from **one specific kind of data**. A chatbot learned from enormous amounts of human text. An image classifier learned from photos with labels attached. A recommender learned from what people like you watched, bought or liked. Different food, different learning.

### AT A GLANCE · THE WHO-LEARNS-WHAT MAP

| The AI product | What it learned from (its data) | What it can do | What it cannot do |
|---|---|---|---|
| chatbot | huge amounts of human writing | continue a conversation, explain, draft | know whether what it wrote is TRUE |
| image classifier | thousands of photos, each with a label | sort new photos into its labels | explain **why**, or see what no photo showed |
| recommender | records what similar people chose | suggest the next video or product | understand you, or ask you a question |

*Keep this map in mind all year. Every AI system is somewhere on it — and the last column is where engineers earn their salary.*

### TWO MACHINES, HONESTLY COMPARED

| Pick one from my morning list | What it learned from | Its honest cannot-do |
|---|---|---|
| | | |
| | | |

*The who-learns-what map in ONE sentence of mine:*

**ENGINEER'S DRILL · SORT THE SIX — WHO LEARNS WHAT?**

Place each product on the map: what did it learn from, and what is its honest cannot-do? Write C (chatbot), I (image classifier) or R (recommender) — then one cannot-do of your own:

| The product | C / I / R? | Its honest cannot-do |
|---|---|---|
| a video app that queues up cricket highlights for you | | |
| a photo app that names the dog breed | | |
| an exam-doubt solver that writes full explanations | | |
| a shop bell that offers coupons based on your old bills | | |
| a scanner that spots cracked railway panels | | |
| an email filter that learns which mail you mark as junk | | |

*Two of these learned from the SAME kind of data — can you spot which pair? That pair is your proof that data, not the product name, decides what a machine can do.*

**QUICK POLL — THE FAMILY QUESTION**

Of the six products above, which ONE would your family actually pay for — and which one would they refuse on principle? Two lines, one reason each:

**PREDICT THE LEARNING** Which ONE product above do you predict learned from the FEWEST examples — and what could go wrong for it? Two lines:

## T8-01 Problem Statement Sharpener

QUICK WIN · PAIR · 15 MIN

Real projects die from **vague problems**. “Our village has a water problem” cannot be built; “detect which public taps run dry for more than two days, so the ward office can dispatch repairs” can. The sharpening tool is the **4W test** — **Who** exactly has the problem? **What** exactly goes wrong? **Where** does it happen? **Why** does it matter? Take these three blunt complaints and sharpen each one into a problem statement a engineer could actually plan around:

| The blunt complaint | Your sharpened problem statement (use all four Ws) |
|---|---|
| “Students lose their notebooks.” | |
| “The bus is always late.” | |
| “People waste water.” | |

**Now the hardest step:** cross out one W in your bus statement and watch it wobble. Which W was load-bearing — remove it and the plan collapses?

WORD POWER · 1

**Definition: problem statement**

One precise sentence saying exactly what problem you solve, for whom, where.

*Read it three times today — it sticks!*

**MY MORNING MACHINES — name three "smart" helpers you met before school:**

1 · _____ 2 · _____ 3 · _____

*By page 20 you will be able to say which of the three learned from examples — and what its data was. Watch your list change.*

**ENGINEER'S NOTE**

A good problem statement sounds boring — and that is a feature. "Reduce" and "improve" are dreams; "detect", "count", "predict X for group Y in place Z" are plans. In the capstone of this book you will write one of these for a real problem you choose, and your classmates will pressure-test it. Sharpen now; shine later.

## 1.2 The loop every AI project runs

Here is the whole subject of this book in four boxes. First, **problem scoping**: turn a vague worry into a sharp statement (you just did this). Second, **data**: find or collect examples the machine can learn from — and check who is inside them. Third, **model building and testing**: train on some examples, test on hidden ones, and measure honestly. Fourth — the step beginners skip — **reflection and improvement**: the first version is always wrong somewhere; the team reads the failures, fixes the data or the design, and loops back. Notice the arrows: this is a **cycle**, not a straight road. Professional teams go around it many times before anything ships.

**SHARPEN TWO WISHES — SCOPE LIKE AN ENGINEER BEFORE THE DIAGRAM**

**Wish 1:** "The school buses are always late." → Your sharp version (who / what / where / why now):

**Wish 2:** "Everyone wastes water." → Your sharp version:

*Notice what sharpening does: it chooses the DATA you would need. That is why scoping comes first in the cycle below.*

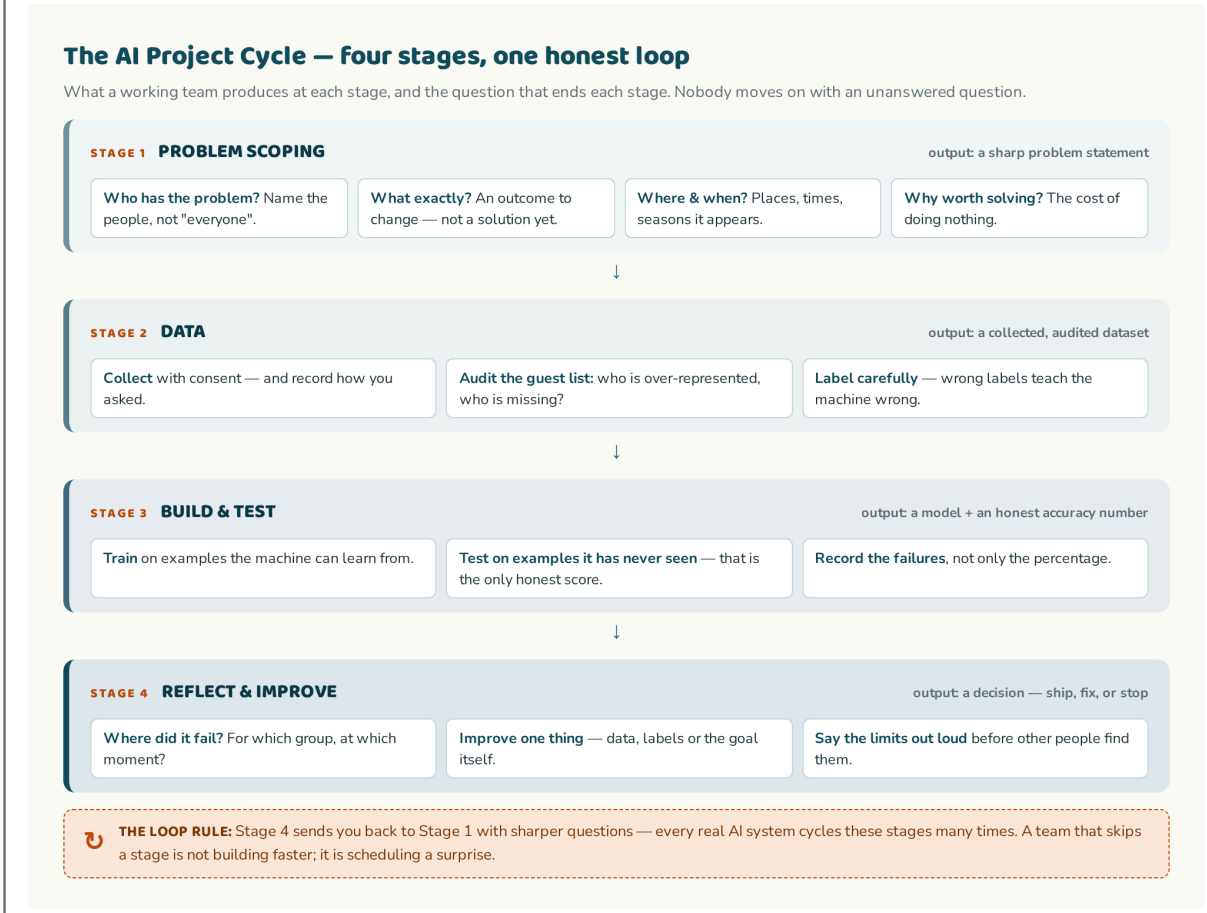



**FIG** The AI project cycle with each stage's output and ending question — the loop that T8-02 sequences and T8-03 walks.

**THE LOOP IN ONE MINUTE** Explain the diagram to a Class 6 student in exactly two sentences — one for the boxes, one for the arrows:

**LOOP SPOTTER — RUN THE CYCLE ON YOUR OWN CLASS 7 SURVEY**

Which of the four stages did your Class 7 survey machine do **worst** — and what would Stage 4 (reflect and improve) have changed first?

## T8-02 Cycle Cards

HANDS-ON LAB · GROUP · 15 MIN

Your teacher has an envelope with five cards: **PROBLEM**, **DATA**, **MODEL**, **TEST**, **IMPROVE**. (No envelope? Copy the five cards onto paper slips.) As a team, lay them out in the order a real project runs, then defend your order — one line per card:

| Card | Position we chose (1st, 2nd…) | Why it sits there — one line |
|---|---|---|
| PROBLEM | | |
| DATA | | |
| MODEL | | |
| TEST | | |
| IMPROVE | | |

**The twist question:** after IMPROVE, which card does the team return to? Draw the loop arrow on your table — and explain in one line why the cycle never really "ends" for a machine that keeps meeting new people:

WORD POWER · 2

**Definition: AI project cycle**

The loop: scope the problem, collect data, build and test, reflect and improve.

*Read it three times today — it sticks!*

WORD POWER · 3

**Definition: dataset**

The collection of examples a machine learns from, or is tested on.

*Read it three times today — it sticks!*

**ENGINEER'S DRILL · NAME THE STAGE**

Each sentence below comes from some AI project. Write which stage it belongs to: P (problem), D (data), M (model), T (test) or I (improve):

| What the team did or said | Stage? |
|---|---|
| "We spent two days deciding: are we predicting **late buses** or **crowded buses**? Pick one." | |
| "We asked 60 commuters to log their wait times for two weeks." | |
| "We taught the machine on 40 logged trips, keeping 20 trips hidden." | |
| "On the hidden trips it was right 13 times out of 20 — let's look at the 7 misses." | |
| "The misses were all rainy-day trips. Version 2 gets a weather column." | |
| "Wait — before any of this: WHO exactly suffers from late buses, and where?" | |

**■ MYTH BUSTER**

**MYTH** *"AI = ChatGPT."*

**TRUTH** A chatbot is one famous resident of the AI city, not the city itself. The spam filter in your mail, the crop-disease scanner in a field office, the queue predictor at a clinic — all AI, none of them chat. And most of them learned their skill from a carefully chosen **dataset** through the very cycle you just sequenced. When someone says "AI", train yourself to ask: **which machine, learned from what data, tested how?** That question is the whole of this book.

## 1.3 The Plant Doctor case: a complete walk-through

Time to watch the whole cycle run — and fail. Read this true-to-life case from the fictional village of Amravati. A team built an AI to diagnose plant disease from leaf photos. Every stage was done with enthusiasm. Your mission is to find where it **actually** went wrong, because the report card said only: "accuracy 82%."

**PREDICT THE FAILURE — BEFORE YOU READ THE DIARY**

An AI project done "with enthusiasm" by one person in one week usually fails at the same stage. Which stage do YOU predict, and what is your evidence from the sentence above?

## T8-03 Plant Doctor Walk-through

GROUP · 25 MIN

Read the project diary, then answer as a team.

**PROJECT DIARY — PLANT DOCTOR v1**

**Stage 1 — Problem:** "Help farmers know if a plant is sick." (The team liked this slogan so much they kept it.) **Stage 2 — Data:** A member collected 2,000 leaf photos in one week — all from **his own farm's hybrid tomato field**, taken at **noon in full sun**, all labelled by **one nephew** who "knows plants". **Stage 3 — Model & Test:** They trained on 1,600 photos and tested on 400 hidden ones. Accuracy: 82%. **Stage 4 — Deploy:** The app was announced at the market. Two weeks later the complaints arrived: it called every diseased leaf "healthy" on the small farms near the river, and it knew nothing about the local desi (native) varieties that look completely different from hybrid tomatoes.

| Diary stage | One thing that was done too fast or too narrowly | What the team should have done instead |
|---|---|---|
| 1 · Problem | | |
| 2 · Data | | |
| 3 · Model & test | | |
| 4 · Reflection | | |

**The accuracy trap:** 82% sounds like a good score. Write two reasons why that single number hid the failure completely:

**Whose problem statement would you write instead? One sharpened sentence for the REAL users:**

## Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can turn a vague complaint into a problem statement with Who, What, Where and Why ○○○
- [ ] I can name the four stages of the AI project cycle and explain why it loops ○○○
- [ ] I can place a chatbot, an image classifier and a recommender on the who-learns-what map ○○○
- [ ] I can point at the Plant Doctor diary and name the stage that failed first ○○○

### ■ THINK & INK

*A "smart" app or service my family uses was really built for someone else. Its problem statement was probably written for ..., because ...*

### ENGINEER'S DRILL · SHARPEN OR SPOIL?

Two problem statements enter the workshop. Mark each S (sharp enough to plan with) or V (still vague), then fix the vague one:

| Problem statement | S or V? | If V — the missing W |
|---|---|---|
| "Predict which water pumps in Ward 12 will fail this monsoon, so repairs happen before the rains." | | |
| "Make education better with AI." | | |
| "Flag Class 8 homework sheets that arrive blank, so the teacher can check 3 instead of 40." | | |
| "Help people." | | |

### CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed building AI starts with the machine. Now I know it starts with ...

**WHERE WOULD YOU ENTER THE LOOP?** A friend's school already has "students forget things". Which cycle stage would YOU enter the loop at — and what would you build your version on?

**CHAPTER 1 CLUES — POCKET THESE**

- **AI** is the field; **machine learning** is the engine that learns patterns from a **dataset**.
- A **problem statement** passes the 4W test: Who, What, Where, Why — all four, or it wobbles.
- The project cycle loops: **problem → data → model & test → reflect & improve → back again**.
- One accuracy number can hide **who** the machine fails — the Plant Doctor proved it.

**The clue I would tell my family tonight:**

## Chapter 1 Checkpoint

one line each — no peeking back!

**ALONE · 3 MIN**

**1.** A complaint says "fix the school toilets." Write its sharpest one-line upgrade using at least three of the four Ws.

**2.** The Plant Doctor team tested on 400 hidden photos and scored 82%. Name the TWO diary stages that deserved the blame anyway.

**3.** A chatbot and an image classifier — which one learned from human writing, and what is the other one's honest cannot-do?

**MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT THE PROJECT CYCLE**

# 2. Data & Fairness

**GUIDING QUESTION**

*If the data only knows some of us, who does the machine actually serve?*

A machine is a mirror of its dataset — show it only some people and it serves only some people. In this chapter you audit real tables like a data auditor (counting who is over-represented and who is missing), run a sampling game that ends with a surprise, repair an unfair dataset three different ways, and write the consent rules every honest collector must follow. This is the chapter where engineers learn that "the data said so" is never the end of the story.

**MISSIONS** T8-04 – T8-07
**OUTCOMES** 8.D1 · 8.D2 · 8.R1 **STRANDS** D R

*Links: Maths — percentages, sampling · Social Science — inclusion & community · English — persuasive asking*

**In this chapter**


2.1 The audit: who is inside the table? ..... 22
2.2 The sampling game: small groups, big surprises ..... 25
2.3 Fairness Fix-It: repairing a broken guest list ..... 27
2.4 Consent: the door every example walks through ..... 30


**TERMS YOU'LL BUILD WITH**

sampling · consent · fairness · representation

**WARM-UP · BEFORE YOU READ** — ALONE · 2 MIN

If a "smart attendance" camera was trained only on morning-shift students, who would it fail to recognise — and why would that matter? Write one line.

**ENGINEER'S TOOLKIT — YOU WILL NEED:** pencil · calculator or rough column · class register (or a made-up list of 24 names) · sticky notes

## 2.1 The audit: who is inside the table?

Every dataset is a guest list. Somebody decided who got invited — which streets, which ages, which languages, which crops — and everybody left off the list simply does not exist **for the machine**. Data scientists call the balance of that guest list **representation**: does the dataset look like the real group of people the system will meet? You already know what happens when examples are few or one-sided (Class 7, T7-05). This year we go further: we **count**. A fair audit does not say "we should be more inclusive" — it says "40 of our 50 examples are from one ward; that ward is 20% of the town; we are overweighted by a factor of four." Numbers turn a feeling into a plan.

**GUEST LIST OF MY CLASSROOM — A 60-SECOND PRACTICE AUDIT**

If a machine were trained only on the students sitting in your row or group right now, which TWO voices or habits from the wider class would it never learn?

**QUICK COMPUTE — WARM UP THE AUDIT MUSCLE**

| If the dataset has… | …and this many are from Ward 3 | What percentage is that? |
|---|---|---|
| 40 examples | 10 | |
| 200 records | 120 | |
| 50 photos | 2 | |

*Same three numbers, three different weights — percentages are how an audit speaks out loud.*

**TWO-MINUTE VOLLEY — SAY IT OUT LOUD WITH A PARTNER**

Take turns, one sentence each: **"A dataset is a guest list because …"** — then **"Consent matters because …"** — then **"A fix without a price is …"** Two lines for the best sentence you heard:

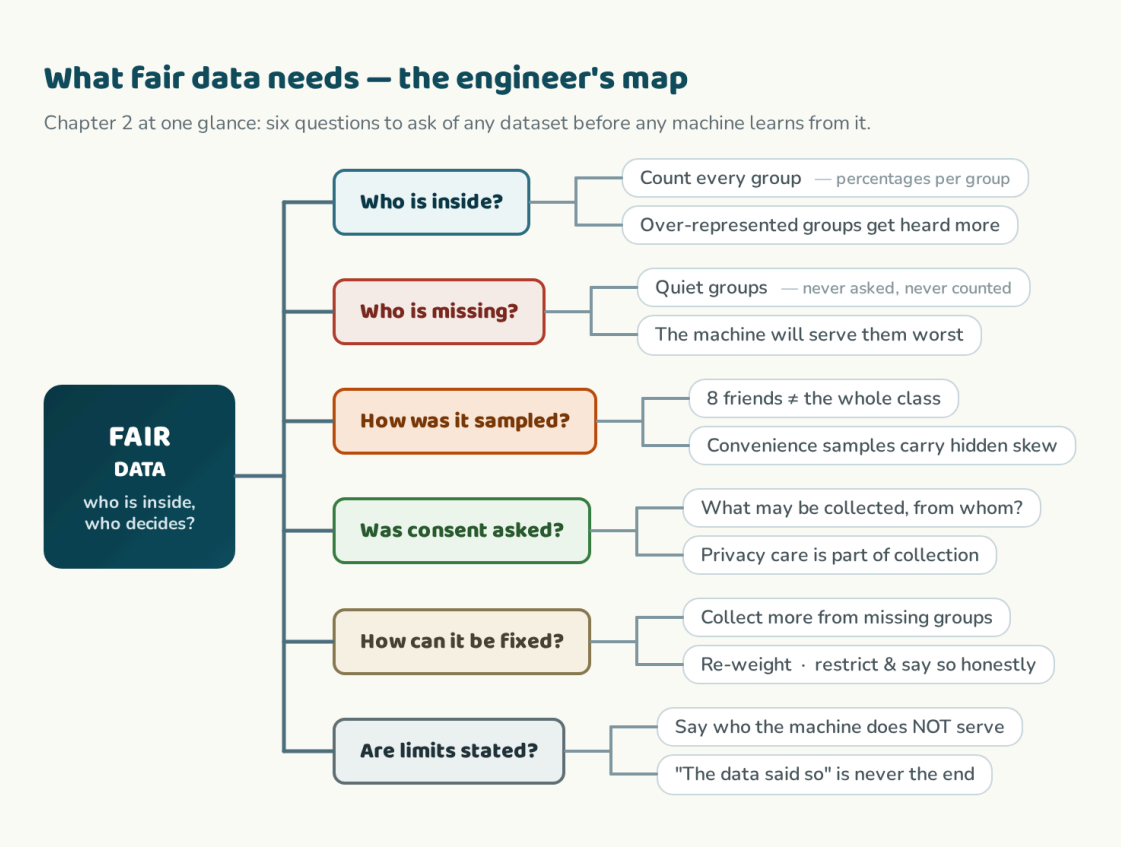



**FIG** The fair-data map: six questions your audits must answer before any machine learns from a dataset.

### WARM-UP AUDIT — THE DATA YOU ALREADY ARE

Your school register is a dataset about you. Answer the map's first two questions about it, honestly, one line each:

**Who is inside?**

**Who is missing — and who decided?**

*Hold your answers — T8-04 hands you a harder table in a moment.*

## T8-04 Data Detective

GROUP · 20 MIN

A city clinic built a “fever triage” helper trained on 200 patient records. Before you audit: **predict first.** Which group do you EXPECT to be overweighted in a clinic’s own records — and why? Write one line, then audit for real:

### THE 200 TRAINING RECORDS

| Group in the records | Records |
|---|---|
| Adults, Ward 3 (clinic’s own street) | 120 |
| Adults, other wards | 30 |
| Children, Ward 3 | 40 |
| Children, other wards | 10 |
| Elderly, any ward | 0 |
| Patients who speak only Kannada | 4 |

### THE TOWN AS IT REALLY IS

| Group | Share of town |
|---|---|
| Adults total | 55% |
| Children total | 30% |
| Elderly total | 15% |
| Ward 3 residents | 18% |
| Kannada-only speakers | 22% |

*Percent to write: records ÷ 200 × 100. A share 2× or more off from reality is a red flag — mark it.*

| Group | % of dataset | Town share | Over / Under / Missing? |
|---|---|---|---|
| Adults, Ward 3 | | 18% | |
| Children, other wards | | 30% total | |
| Elderly | | 15% | |
| Kannada-only speakers | | 22% | |

**The verdict in one sentence — who will this machine serve best, and who will it fail?**

**MY AUDIT HABIT — the first real place I will use percentages this month:**

I will audit … _____ because its guest list matters to …

**WHO DECIDED? — the question behind every guest list:** For the clinic dataset, one person quietly decided who got invited. Name the two people who SHOULD have been in that room:

**WHY NOT COLLECT EVERYONE?** Name one real reason a project cannot survey every person it serves — and what that forces the team to become good at:

**WORD POWER · 4**

Definition: **sampling**

Choosing part of a group to study, hoping it fairly represents the whole.

*Read it three times today — it sticks!*

## 2.2 The sampling game: small groups, big surprises

Why not just collect **everyone**? Cost, time and energy force every real project to sample — to study a part and treat it as a stand-in for the whole. The danger lives in **how** the part is chosen. Pick your sample from the people nearest to you, and the sample quietly inherits all your neighbourhood's habits. Pick it at random from the whole group, and the surprises begin to even out. You are about to feel this in one round of a game — the class meeting 8 students will **not** look like the class.

**PREDICT THE GAME — BEFORE YOU PLAY IT**

The first 8 hands will lean toward which kind of student? And which kind of student will the bag draw fairly? Write both predictions — we check them against the real numbers soon:

**A sample I already trust (and why):** _____ **— one thing it still misses:**

## T8-05 Sampling Game

HANDS-ON LAB · CLASS · 15 MIN

Round 1: your teacher picks the **first 8 students** who raise a hand (or the front row). Each writes one number: **minutes spent on screens yesterday**. Record them, then find the class's average for the **whole class** (your teacher writes it on the board without revealing it until Round 3).

| What we measured | First-8 sample | Whole class (revealed last) |
|---|---|---|
| Screen minutes — average | | |
| How far the sample average was from the truth | | |

Round 2: now sample 8 **fairly** — assign everyone a number and draw 8 from a bag. Record the new average and the new gap:

| What we measured | Fair random 8 | Gap from truth |
|---|---|---|
| Screen minutes — average | | |

**Which sample landed closer to the truth — and why was the hand-raising sample biased before anyone wrote a single number?**

**A engineer collected training photos only from her own school. Which round of this game is she stuck in?**

WORD POWER · 5

Definition: **representation**

How closely a dataset's people match the real people the system will serve.

*Read it three times today — it sticks!*

**ENGINEER'S DRILL · SPOT THE SKEWED SAMPLE**

Each plan collects data for a school-canteen menu predictor. Mark each sampling plan F (fair) or S (skewed), and name the group it would miss:

| The sampling plan | F or S? | Who is missing |
|---|---|---|
| Ask the 12 students in the robotics club. | | |
| Put a slip in every lunch bag for one week; use whoever replies. | | |
| Number all 240 students; draw 60 from a box. | | |
| Survey only students who buy canteen food. | | |
| Survey at 8:00 am only — bus students have not arrived yet. | | |

*Trickiest: the lunch-bag plan **sounds** fair — but it only hears from families that pack lunch and from students willing to answer. "Random-ish" is not random.*

## 2.3 Fairness Fix-It: repairing a broken guest list

Finding the skew is audit work; **fixing** it is engineer work — and there is never one magic fix. Real teams choose among several repairs, each with a price. Collect **more** data from the missing groups (slow but strongest). **Re-weight** what you have (fast, but the thin groups stay thin). **Restrict** the machine to the groups it truly knows, and say so honestly on the label (safest, but less useful). Notice what is **not** on the list: pretending the machine is neutral. A machine trained on a skewed guest list is not neutral — it votes for the majority group every single time.

**SAMPLE-SIZE INTUITION — BEFORE YOU CHOOSE A FIX**

A re-balance can divide by ten, but it cannot invent examples. In one line: what is the DIFFERENCE between hearing a group 4 times instead of 40 times?

### ENGINEER'S DRILL · WHICH FIX WOULD YOU DIAL?

Each team has found its skew and must dial one fix — C (collect more), R (re-balance what exists) or L (limit the machine and label it honestly). Choose, and write the price:

| The situation | C / R / L? | The price in one line |
|---|---|---|
| The fair-weather bus data can be collected — the monsoon is two months away. | | |
| The exam is on Friday; there is no time to gather new examples. | | |
| The clinic cannot reach elderly patients safely this season. | | |
| New data exists, but only if the school pays for a second survey round. | | |

*There is no universally right dial — a fix is right when its price is one the team can honestly pay. That is a engineer's judgement, not a maths answer.*

### PRICE CHECK — WHAT DOES A FIX ACTUALLY COST?

| The fix | Weeks of work (our estimate) | The group it helps first |
|---|---|---|
| Collect 40 new elderly records (home visits) | | |
| Re-balance the existing 200 records | | |
| Restrict to adults + print an honest label | | |

### MY NEIGHBOURHOOD GUEST LIST — where do OUR examples come from?

If someone collected "how people cross the road" only near our school gate, which crossing habit of the wider neighbourhood would they completely miss?

## T8-06 Fairness Fix-It

GROUP · 20 MIN

Return to the clinic's fever helper (T8-04): zero elderly records, 2% Kannada-only records. Your team must propose **three different fixes**, one of each kind, and be honest about the price of each:

| Fix | Exactly what we would do | The price we pay (time / cost / usefulness) |
|---|---|---|
| 1 · Collect | | |
| 2 · Re-weight or re-balance | | |
| 3 · Restrict honestly | | |

**Which fix would you ship first, and what would you tell the ward office about what the machine still cannot do?**

WORD POWER · 6

Definition: **fairness**

A system works equally well for every group of people it affects.

*Read it three times today — it sticks!*

■ MYTH BUSTER

**MYTH** *"Fairness means treating everyone the same."*

**TRUTH** Same treatment is only fair when everyone started in the same situation. The clinic helper that scores every patient against adult, Ward-3 patterns treats everyone "the same" — and that is exactly the unfairness. Real fairness asks a harder question: **does the system work equally well for every group it touches?** Sometimes fair treatment means collecting **different** extra data for different groups — more elder-friendly examples, more Kannada-language cases — so the machine's **outcomes**, not just its rulebook, come out level. Fairness is measured in results, not in uniformity.

## 2.4 Consent: the door every example walks through

Behind every row of a dataset stands a person who breathed, spoke, walked or was photographed — and every honest project asks before it takes. **Consent** means permission given freely, by someone who understands what will be collected, why, and who will see it. Children and other vulnerable groups deserve extra care: often a parent or guardian must say yes alongside them. And consent has a shy cousin called **privacy care**: even data given happily must be stored safely, used only for the promised purpose, and stripped of details that could identify someone — a name, an address, a face. Engineers who skip this step do not just break rules; they break the trust their whole project runs on.

### CONSENT — TRUE OR FALSE?

| The claim | T or F? | The one-line why |
|---|---|---|
| “They said yes quickly, so we do not need to explain the project.” | | |
| “Blurring faces afterwards can fix consent that was never asked.” | | |
| “Data collected for the canteen survey may be reused for attendance.” | | |

### OUR CONSENT PROMISE — PRACTICE COPY

Draft the one sentence you would say to the class BEFORE the corridor monitor collects anything. The best drafts name what, why, who sees it, and how to say no:

**Self-grade your draft:** names WHAT is collected ☐ · says WHY ☐ · says WHO sees it ☐ · offers a NO path ☐

### WHO SEES IT? — THE FORGOTTEN CONSENT QUESTION

| If the noise monitor's numbers went to… | My comfort (1–3) | What I would demand first |
|---|---|---|
| …only the class teacher’s register | | |
| …a wall chart everyone can read | | |
| …the school’s app for parents | | |

*Comfort levels are data too — and they are exactly the data most projects never collect.*

## T8-07 Consent Check

PAIR · 10 MIN

A school AI club wants to build a “corridor noise monitor” that records sound levels to suggest quieter break times. For each data idea below, mark: G (fine to collect with plain consent), A (needs **extra** care — guardian consent, or faces/voices anonymised), or N (should not be collected at all). Then write the team’s one-sentence consent promise:

| Data the club wants | G / A / N? | Why — one line |
|---|---|---|
| Sound **level** numbers (no words, no voices) | | |
| Actual voice recordings from the corridor | | |
| Photos of students’ faces while noisy | | |
| Blurred silhouettes with noise numbers attached | | |
| Names of the loudest five students, published on a wall | | |

**Our consent promise (one sentence we would say to every student before collecting anything):**

WORD POWER · 7

Definition: **consent**

Permission given freely by a person who understands what will be collected and why.

*Read it three times today — it sticks!*

WORD POWER · 8

Definition: **privacy**

Keeping people’s personal details safe, used only for the promised purpose.

*Read it three times today — it sticks!*

## Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can audit a dataset with percentages and name the groups that are missing or overweighted ○○○
- [ ] I can explain why a hand-raised sample is biased before anyone answers a question ○○○
- [ ] I can propose three kinds of fairness fixes — and state the price of each ○○○
- [ ] I can decide what data needs plain consent, extra care, or no collection at all ○○○

### THINK & INK

*One machine or service I know was surely built on a skewed guest list. The group it was built without is …, and one thing that changes for them is …*

### PERCENT PRACTICE — FINISH THE AUDIT

The clinic audit in numbers — complete each row (records ÷ 200 × 100):

| Group | Records | % of 200 | Verdict |
|---|---|---|---|
| Adults, Ward 3 | 120 | | |
| Children, other wards | 10 | | |
| Kannada-only speakers | 4 | | |

*Verdict vocabulary: **overweighted** (far above the town share) · **underweighted** (far below) · **missing** (zero or nearly zero). A group cannot argue with its own percentage.*

**The one audit finding I will remember longest (group, number, feeling):**

**HOME LINK** *take this page home — no screen needed*

## AT HOME Guest List Audit

HOME · 15 MIN

With an adult, pick one app or service your family uses (a video app, a map, a shopping site). Do not open any settings or accounts — just talk. Guess together: whose voices, faces, languages and places were probably in its training data — and who might be missing?

**Our guess at the guest list (who is inside):**

**Who is probably missing — and one way that could go wrong for them:**

**One question we would ask the makers if we met them:**

**CASE NOTES — WHAT CHANGED IN MY THINKING?**

Before this chapter I believed data was just… data — neutral stuff. Now I know every dataset has a guest list, and the auditor's first question is always …

**CHAPTER 2 CLUES — POCKET THESE**

- A dataset is a **guest list** — first ask who is inside, in percentages, not feelings.
- Hand-picked samples inherit the picker's neighbourhood; **random** samples land nearer the truth.
- Three honest fixes for a skewed dataset: **collect more, re-balance, or restrict and say so**.
- **Consent** is free, informed permission — and privacy means keeping the promise after it is given.

**The clue I would tell my family tonight:**

**MY SAMPLE, AUDITED BY ME — the honest version:**

My own T8-05 sample leaned toward … _____ · the fair version would need …

## Chapter 2 Checkpoint

one line each — no peeking back!

**ALONE · 3 MIN**

**1.** The clinic dataset had 120 adult Ward-3 records out of 200. What percentage is that — and why is it a red flag against an 18% town share?

**2.** Name the three fixes for an unfair dataset, and the price of the one you would choose.

**3.** Why is "the machine treats everyone the same" NOT a proof of fairness?

**MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT DATA, FAIRNESS AND CONSENT**

# 3. How Models Learn & What No-Code Tools Do

**GUIDING QUESTION**

*How does a machine decide — and how do we measure whether it decides well?*

Now the workshop floor. You will hand-run the simplest learning machine there is — a nearest-example classifier that judges new cases by their most similar labelled examples — and then grade it with a real accuracy percentage on a hidden test set. After that you step back and survey the no-code tool market: chatbots, image classifiers and data predictors, each with inputs, outputs and hard limits. By the end you will know which tool fits which problem — and which tool marketing would love you to buy for the wrong job.

**MISSIONS** T8-08 – T8-10
**OUTCOMES** 8.L2 · 8.W1 **STRANDS** L W

*Links: Maths — distance on a grid, percentages · Science — leaf features · Social Science — honest advertising*

**In this chapter**


3.1 The nearest-example machine ..... 36
3.2 The honest score: accuracy on a hidden test ..... 38
3.3 The Paper Classifier Lab: build, test, fail, improve ..... 42


**TERMS YOU'LL BUILD WITH**

nearest neighbour · accuracy · no-code tool · classifier

**WARM-UP · BEFORE YOU READ** — ALONE · 2 MIN

Your new neighbour asks "am I like the family on the left or the right?" — how would YOU decide? Write the one feature you would look at first.

**ENGINEER'S TOOLKIT — YOU WILL NEED:** pencil · ruler · calculator or rough column · 10 blank slips for test cards

## 3.1 The nearest-example machine

Here is the simplest honest answer to "how does a machine decide?" — **it compares**. A nearest-example classifier stores labelled examples, and when a new case arrives it finds the most similar examples it has seen and borrows their labels. That is all. No magic, no understanding — just similarity. On a grid, "similar" has a precise meaning you can measure with a ruler: the **shortest distance**. Two dots close together are two similar cases; the new point takes the label of whichever class lives nearest. You ran a cousin of this machine in Class 7; today you run the real one, with distances and a score.

**RULER WARM-UP · WHICH DOT LIVES NEAREST?**

A new bird sits at (5, 5). Three labelled dots live nearby: P at (4, 6), Q at (7, 5) and R at (5, 2). For each pair of dots, which is closer to the new bird — and what label would the machine vote if P and R are POND birds and Q is a FOREST bird?

| Compare | Nearer to (5, 5)? | So the machine leans… |
|---|---|---|
| P (4, 6) vs Q (7, 5) | | |
| Q (7, 5) vs R (5, 2) | | |
| P (4, 6) vs R (5, 2) | | |

*Ruler trick: count grid squares left-right and up-down, then compare. Your eyes are good at this — trust the ruler more than the vibes.*

**MY RULER METHOD — how I will decide "nearest" in T8-08**

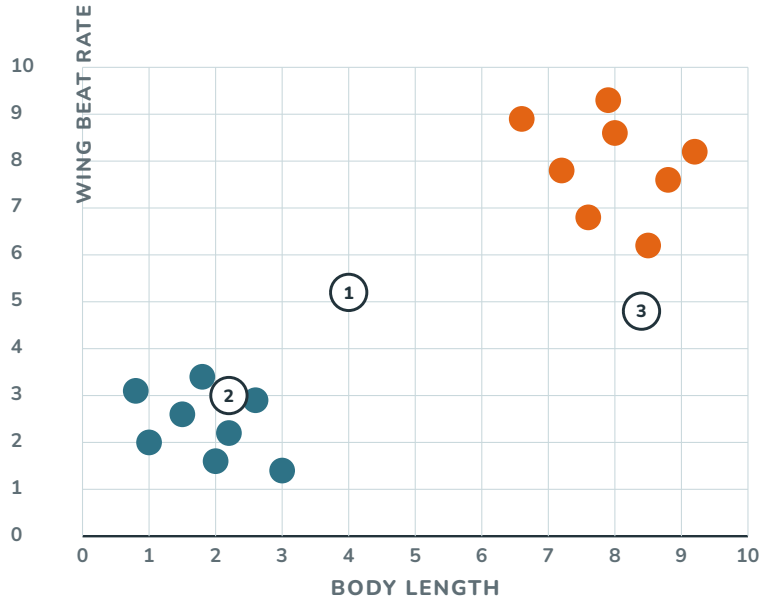
In one line: I will find the nearest dot by … _____ and if two dots are almost tied I will …

**WHICH FEATURE DECIDES?** A bird sits halfway between the pond group and the forest group. Body length says FOREST; wing-beat says POND. In one line: what should the machine (and the birder) do?

## T8-08 Nearest Neighbour by Hand

HANDS-ON LAB · PAIR · 20 MIN

The plot shows a bird-watching classifier's training examples. **Teal dots** = examples labelled POND bird; **amber dots** = examples labelled FOREST bird. The x-axis is **body length (units)**, the y-axis is **wing beat rate (units)**. Three new birds arrive — the hollow numbered circles. For each, use a ruler: find the nearest teal or amber dot, and assign the label.




| New bird | Nearest dot (label + its position) | My classification |
|---|---|---|
| 1 (4.0, 5.2) | | |
| 2 (2.2, 3.0) | | |
| 3 (8.4, 4.8) | | |

**Bird 3 is the interesting one:** it sits near the middle — long body like a forest bird, low wing-beat like a pond bird. **Which feature decided its case, and what would you tell the app's users about cases like this?**

WORD POWER · 9

**Definition: nearest neighbour**

Classifying a new example by its most similar labelled examples.

*Read it three times today — it sticks!*

Definition: **accuracy**

The share of test examples a machine gets right, written as a percentage.

*Read it three times today — it sticks!*

## 3.2 The honest score: accuracy on a hidden test

You measured accuracy informally last year; now it gets a formula a engineer can defend. **Accuracy = correct predictions ÷ total test cases × 100%.** Two rules make the number honest. First, the test set must be **hidden** during training — a machine graded on its own practice questions tells you nothing (you proved this in Class 7's Practice vs Exam). Second, one overall percentage can hide **which** cases fail — so serious engineers also count failures **per group**. Accuracy answers "how often is it right?"; it never answers "right for whom?".

### TRUE OR FALSE — ACCURACY EDITION

| The claim | T or F? | The one-line why |
|---|---|---|
| A machine tested on its own training examples gives an honest accuracy. | | |
| 95% accuracy means the machine is right for every group it serves. | | |
| A bigger hidden test set makes the accuracy number steadier. | | |
| A machine can be accurate on sunny photos and useless at night. | | |

*The two honesty rules in my own words — rule 1 (hidden test):*

*rule 2 (count failures per group):*

### PER-GROUP ACCURACY — WHAT THE ONE NUMBER HIDES

A mango scanner was tested on 100 photos: 60 sunny, 40 shaded. It scored 58 sunny hits and 8 shaded hits. Finish the arithmetic, then answer:

| Group | Correct | Test cases | Group accuracy |
|---|---|---|---|
| Sunny photos | 58 | 60 | |
| Shaded photos | 8 | 40 | |
| Overall — the number the vendor will advertise | 66 | 100 | |

**What does the 66% hide that the vendor will never say out loud?**

**Which group pays the price for that advertising?**

**Write the honest label that belongs on the scanner's box:**

### THE ACCURACY CONTRACT — BUYER AND ENGINEER, ONE SENTENCE EACH

**The buyer will ask:** "accuracy on WHICH photos, counted HOW?" — **the engineer will answer honestly:**

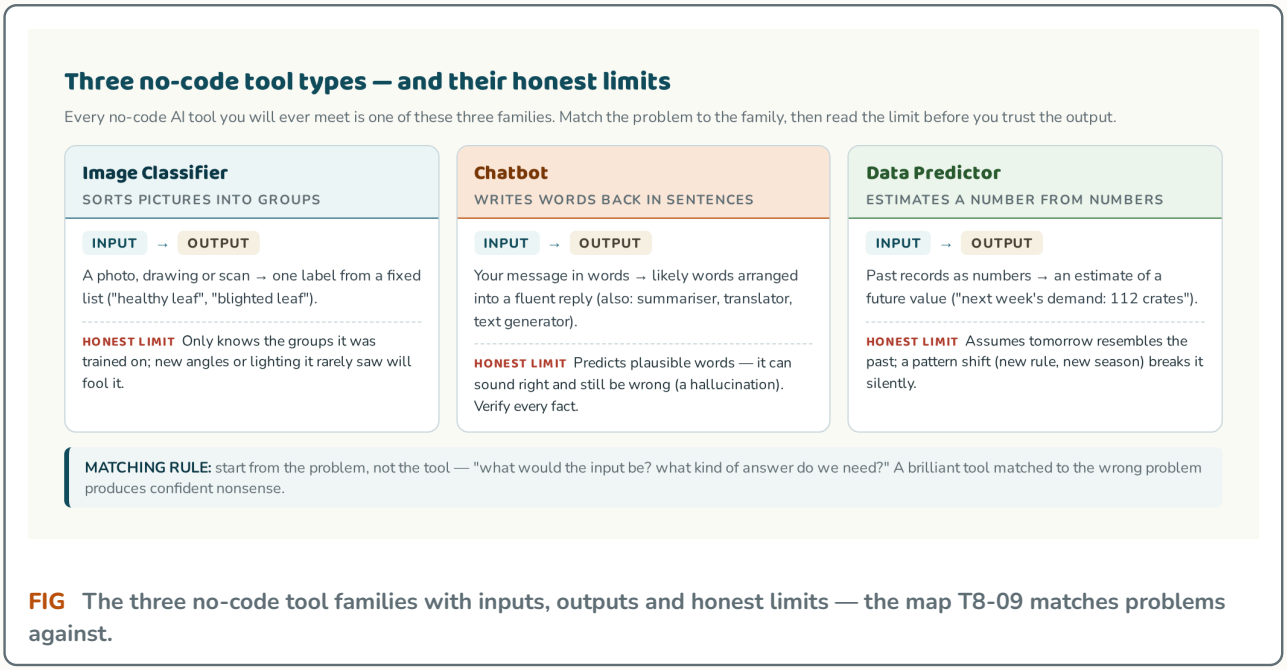



**FIG** The three no-code tool families with inputs, outputs and honest limits — the map T8-09 matches problems against.

## THIRTY-SECOND WARM-UP — NAME THE FAMILY

| Product | Family (classifier / chatbot / predictor) | The one input it must be given |
|---|---|---|
| an app grading leaf photos as healthy or blighted | | |
| a bot drafting your leave application | | |
| a tool estimating Friday's canteen demand in plates | | |

**Your turn — one product from my own week:**

| Product | Family | The one input it must be given |
|---|---|---|
| | | |

## THE MATCHING RULE, IN MY WORDS

Complete it once, from memory — no peeking at the strip above: "Start from the problem, not the tool, because …"

## T8-09 Tool Match

PAIR · 10 MIN

Match each problem to its no-code tool type — image classifier, chatbot, or data predictor — and write its likely **input**, **output**, and one hard **limit**. The first row is started for you.

| The problem | Tool type | Input | Output | One hard limit |
|---|---|---|---|---|
| Sort mangoes as ripe or unripe at the depot | image classifier | photo of mango | ripe / unripe | needs good lighting; no reasons given |
| Answer parents' questions about school timings | | | | |
| Predict next week's canteen rice demand | | | | |
| Spot cracked solar panels from drone photos | | | | |
| Guess which students may need homework help | | | | |

**The trick question:** a shopkeeper says "one chatbot can do all five jobs — it's magic!" Write your two-sentence reply as the classroom's honest tool advisor:

WORD POWER · 11

Definition: **no-code tool**

A ready-made AI tool you train or use without writing any program.

*Read it three times today — it sticks!*

**ENGINEER'S DRILL · WHICH TOOL — AND WHAT COULD GO WRONG?**

For each promise from a (fictional) vendor, write the tool type and the sharpest thing a **Question-Habit** customer should ask before buying:

| The vendor's promise | Tool type | The question I would ask |
|---|---|---|
| "This app looks at a leaf photo and names the disease!" | | |
| "This chatbot answers any exam doubt, always correct!" | | |
| "This predictor knows exactly how many idlis Sunday needs!" | | |

*Remember the last column of the who-learns-what map: every tool's limit lives where its learning ended.*

## 3.3 The Paper Classifier Lab: build, test, fail, improve

Time for the full experience in miniature. In one mission you will train a classifier, test it honestly, watch it fail, and fix it — the whole cycle in 25 minutes. This is the mission professional teams never skip, because **the failures are the syllabus**. Your lab: two leaf types from the school garden (or from the cards your teacher provides), a feature table, and a test set you must not peek at.

**FEATURE PRE-THINK — 60 SECONDS BEFORE YOU COLLECT**

The two features I plan to measure are ... and ..., because I expect them to separate the leaf types better than colour would.

**LAB BENCH — WRITE YOUR ROLES BEFORE YOU TOUCH A LEAF:** Collector: _____
Scribe: _____

Ruler: _____ Judge (keeps the test set honest): _____

**SAFETY AT THE BENCH:** collect only fallen leaves — never strip a living plant · wash hands after the garden · the Judge hides the test set BEFORE training begins, not after.

## T8-10 Paper Classifier Lab

HANDS-ON LAB GROUP · 25 MIN

Work as a lab team of four. Roles: **Collector** (gathers leaves), **Scribe** (fills the table), **Ruler** (measures features), **Judge** (hides the test set and keeps score).

**Step 1 — Collect.** Gather 6 leaves of type A (e.g., neem) and 6 of type B (e.g., hibiscus). The Judge secretly keeps 2 A-leaves and 2 B-leaves aside as the **test set**. **Step 2 — Train.** For the 8 training leaves, fill the feature table and circle the feature(s) that separate the types best.

| Leaf | Edge: smooth / toothed? | Length (cm) | Label A or B? |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |

**Step 3 — Write your rule.** "A leaf is type B when …"

**Step 4 — Test.** The Judge brings out the 4 hidden leaves. Apply your rule, then have the Judge reveal the true labels.

| Hidden leaf | My rule's answer | True label | Right? |
|---|---|---|---|
| T1 | | | |
| T2 | | | |
| T3 | | | |
| T4 | | | |

**My accuracy:** _____ ÷ 4 = _____ %

**Step 5 — Improve.** Name the one thing you would change in version 2 (more examples? a second feature? a measured threshold instead of a vague word?):

## Definition: classifier

A machine that sorts examples into named groups, like ripe or unripe.

*Read it three times today — it sticks!*

### MYTH BUSTER

**MYTH** *"No-code means no thinking."*

**TRUTH** No-code removes the **typing**, never the **judgement**. Somebody still scopes the problem, chooses and checks the dataset, decides which examples teach the tool, picks the honest test, and reads the failures. The Plant Doctor team (Chapter 1) needed no code either — and their project still failed on data. A no-code tool in careless hands is a fast way to build the wrong thing confidently. The thinking you are practising in this book — audit, test, improve — is exactly the part no software can do for you.

### Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can classify a new point by nearest labelled example — with a ruler and a reason ○○○
- [ ] I can compute accuracy on a hidden test set as correct ÷ total × 100% ○○○
- [ ] I can match a problem to classifier, chatbot or predictor, with input, output and one limit ○○○
- [ ] I can run a train → test → improve loop and name what version 2 changes ○○○

### THINK & INK

*The nearest-example machine is basically "judge by closest match". One situation where that habit would be UNFAIR to a person is …, because …*

### ENGINEER'S DRILL · ACCURACY ARITHMETIC

Compute each accuracy, then answer the honesty question below:

| The hidden test | Correct | Total | Accuracy % |
|---|---|---|---|
| Bird classifier, pond vs forest | 17 | 20 | |
| Mango ripeness scanner | 94 | 100 | |
| Cracked-panel spotter | 9 | 10 | |

**But wait** — the cracked-panel spotter's 10 test cases were ALL sunny-day photos. Its real job includes night patrols. What would you tell the buyer about that 90%?

### CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed a machine decides by rules somebody typed. Now I know the nearest-example machine actually decides by …

### CHAPTER 3 CLUES — POCKET THESE

- A nearest-example classifier **compares distances** — similarity, not rules, does the deciding.
- **Accuracy** = correct ÷ total × 100% — and one overall number can hide which groups fail.
- Classifier, chatbot, predictor: each has inputs, outputs and a **hard limit** where its data ran out.
- No-code removes the typing, never the thinking — audit, test and improve are still your job.

**The clue I would tell my family tonight:**

**MY LAB REPORT CARD — THE LOOP IN ONE SMALL BOX**

| Version | My accuracy | The change I made, in one line |
|---|---|---|
| v1 (first rule) | | — |
| v2 (after the fail) | | |
| v2 target if we collected 8 more leaves | | |

*Teams that write the loop down improve faster than teams that merely feel it. Three rows, three honest numbers — that is a engineer's diary.*

## Chapter 3 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

**1.** New point at (3.5, 6.0): nearest teal dot is at (3.0, 5.4), nearest amber dot at (6.2, 6.4). Which label — and what measurement decided it?

**2.** A test set has 25 cases; the machine got 21 right. Its accuracy? Show the division.

**3.** A vendor says his chatbot "solved" mango ripeness. What is the right tool for that job, and what can the chatbot NOT see?

**MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT HOW MODELS LEARN**

# 4. Ethics & Responsible AI

**GUIDING QUESTION**

*Why does a machine that sounds so right sometimes get it so wrong — and who answers for that?*

Meet the smoothest talker in the AI city: the text generator. You will play its game yourself — predicting next words from patterns — and discover why the very trick that makes it fluent also makes it **invent**. Then you hunt hallucinations in a real-looking AI paragraph, take a scenario to court as judge, lawyer and witness, and write your own AI-use pledge. This chapter has no magic answers; it has something better — a routine you can run on every confident machine you will ever meet.

**MISSIONS** T8-11 – T8-14
**OUTCOMES** 8.W2 · 8.R2 **STRANDS** R W

*Links: English — language patterns & argument · Social Science — privacy & public impact · Maths — likelihood*

### In this chapter


4.1 The next-word machine: how text generators actually work ..... 48
4.2 Hallucination Hunt: verify before you trust ..... 50
4.3 The Responsible-Use Court: decisions with consequences ..... 53
4.4 My AI-Use Pledge ..... 54


**TERMS YOU'LL BUILD WITH**

**text generator** **hallucination** **misinformation** **responsible use**

**WARM-UP · BEFORE YOU READ** ALONE · 2 MIN

Finish this sentence the way a friend probably would: "The early bird catches the …". Now finish: "The early crow catches the …". Which one were you surer about — and where did that knowledge come from?

**ENGINEER'S TOOLKIT — YOU WILL NEED:** pencil · the printed paragraph in T8-12 · sticky notes for court roles · an honest heart

## 4.1 The next-word machine: how text generators actually work

Forget magic. A text generator does one thing, enormously well: it reads the words so far and predicts a **likely next word** — based on patterns in a mountain of human writing. It is autocomplete that grew up. The catch is built into the trick itself: the machine knows which words **usually follow**, not which words are **true**. "The early bird catches the worm" flows because millions of people wrote it; a machine completing "The early crow catches the…" will pick whatever sounds plausible. Usually it sounds right. Sometimes it is completely invented — with perfect confidence either way, because confidence and truth live in different rooms.

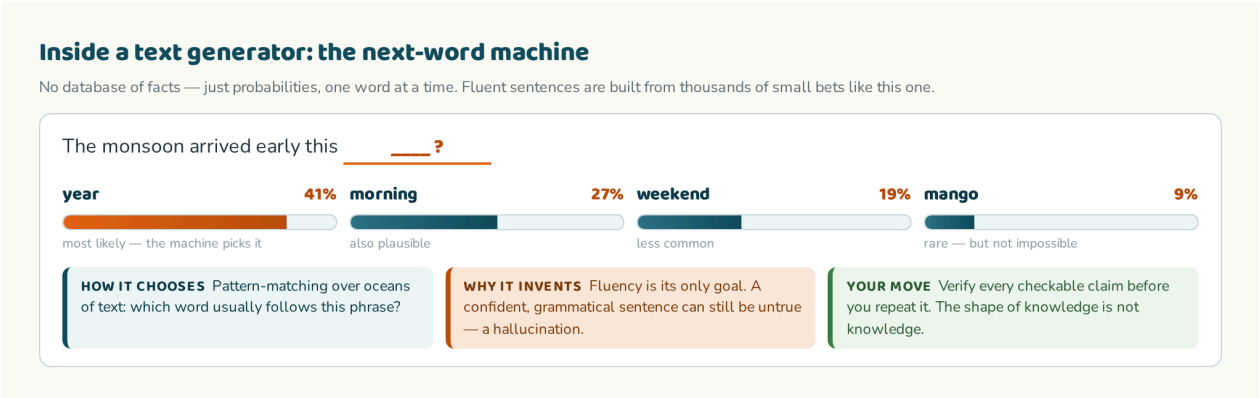



**FIG** The next-word machine at work: one sentence, four candidates, probability bars — fluency without a fact-checker.

**THE FLUENCY TRAP — WHICH SENTENCE DID A MACHINE WRITE?**

**A.** "The RMSC survey of 2,340 households across four monsoon seasons found a 17% drop in borewell reliance — the report is on the department portal, table 6."**B.** "Many experts widely believe that water levels are dropping significantly in various regions, which is concerning for the future."

My call, and the ONE feature that decided it (dates? numbers? sources? hedging?):

## T8-11 Next-Word Game

HANDS-ON LAB · CLASS · 15 MIN

You are the machine. Your teacher reads each sentence aloud up to the blank; your class votes on the most likely next word. Then your teacher reveals the counts from a 500-page book of local writing (invented for this game). Fill the table as you play:

| Sentence up to the blank | Class's first guess | Runner-up guess | Revealed: which word won, and by how much |
|---|---|---|---|
| "The monsoon arrived early, bringing heavy …" | | | |
| "She opened the lunch box and found her favourite …" | | | |
| "The cricket match was delayed because of …" | | | |
| "The old elephant walked slowly toward the …" | | | |

**Now the deep question:** when the machine picked a word nobody in the room expected, was it **wrong**? What was it actually optimising for?

**A classmate asks: "if it's just picking likely words, why do its essays sound so intelligent?" Answer in one line:**

WORD POWER · 13

Definition: **text generator**

A machine that writes by predicting the most likely next words.

*Read it three times today — it sticks!*

**ENGINEER'S NOTE**

Here is why the machine **sounds** smart: likely words, arranged by likely patterns, produce paragraphs shaped exactly like the ones people write. It is a brilliant imitation of the **shape** of knowledge. But shape is not truth. A hallucinated biography of a fake scientist has perfect grammar, a plausible structure and a confident tone — all borrowed from real writing about real scientists. That is why your eyes and your verification routine are still the most important parts of the system.

## 4.2 Hallucination Hunt: verify before you trust

When a text generator states something false as if it were fact, engineers call it a **hallucination** — and it is not a bug that will be patched away next month; it is the direct result of predicting likely words instead of checking facts. So the responsible user carries a routine, not a hope. The Class 6 habit — **check before believing** — now grows up into three moves: **trace the claim** (is there a real source?), **test the number** (do two sources agree?), and **smell the story** (is it suspiciously exactly what I wanted to hear?).

**SAME SHAPE, DIFFERENT TRUTH — A PRE-HUNT DRILL**

Both sentences below are grammatical, confident and plausible. Only one survives a fact-check. Mark which — and name the exact words that made the other one suspicious:

| The sentence | V (verifiable) or S (suspect)? | The words that gave it away |
|---|---|---|
| "The Nagpur district library opens at 8 am on weekdays." | | |
| "Most people agree the town's founder was a poet-king in 1238." | | |

*Watch the give-away words: **most people agree**, **everyone knows**, **it is said that** — confidence borrowed from a crowd nobody can name.*

**One suspicious sentence of my own, to trade with a partner:**

## T8-12 Hallucination Hunt

GROUP · 15 MIN

Below is a paragraph a chatbot produced when a student asked about their town. Hunt it: mark each claim V (verifiable and checks out), F (false — contradicted by the fact box), or U (unverifiable — no way to check from here). Then write the chatbot a revision order.

### WHAT THE CHATBOT WROTE

"Sunapur was founded in 1847 by the poet R. Krishnamurthy, who also invented the sunflower oil press used across the district today. The town's famous three-day Mango Festival, started in 1952, attracts over 5 lakh visitors every year. The Government Higher Primary School, established 1961, was the first school in the state to teach computer science. The town bus route 14 runs every ten minutes from the railway station to the fort."

| Claim | V / F / U |
|---|---|
| Founded in 1847 by R. Krishnamurthy | |
| He invented the sunflower oil press | |
| Mango Festival began 1952, 5 lakh visitors | |
| School was FIRST in the state to teach computing | |
| Bus route 14 every ten minutes to the fort | |

### THE FACT BOX (from the town library)

· Sunapur was founded in **1884**; its founder was a merchant, not a poet. · The oil press story is **unknown to any local record**. · The Mango Festival began in **1972**; visitor numbers are **not published anywhere**. · The school dates from 1961 — true — but the first state school to teach computing was in another district. · Route 14 runs **hourly**, and the town has no fort.

**Notice:** every invented claim was **shaped** like a true one — a date, a founder, a festival. The machine matched the pattern, not the facts.

**My revision order to the chatbot (which claims to fix, and what evidence I demand):**

WORD POWER · 14

Definition: **hallucination**

When a text generator states something false as if it were fact.

*Read it three times today — it sticks!*

Definition: **misinformation**

False content that spreads — by mistake or on purpose.

*Read it three times today — it sticks!*

### ENGINEER'S DRILL · THE THREE-MOVE ROUTINE

Name the move: T (trace the claim), N (test the number) or S (smell the story):

| What the responsible user does… | Which move? |
|---|---|
| Searches for a second, independent source that says the same thing. | |
| Asks: "who gains if I believe this instantly?" | |
| Opens the original article the chatbot **claims** to summarise. | |
| Checks whether "92% of doctors agree" names WHO counted and HOW many. | |
| Feels a thrill of confirmation — and slows down on purpose. | |

### ■ MYTH BUSTER

**MYTH** *"If it is trained, it is true."*

**TRUTH** Trained means the machine found patterns in its examples — nothing more. The Plant Doctor was trained and still failed river-farm tomatoes; text generators are trained on human writing and still invent facts, because their job was never truth-finding, it was pattern-continuing. Training makes a machine **consistent** with its examples. Only **testing against the world** — hidden test sets for classifiers, source-checking for text — makes it trustworthy. Whenever you hear "our AI is trained on millions of examples", respond like a engineer: **trained, yes — but tested how, and on whom?**

**HUMAN VS MACHINE** **The two essays — who wrote which?** HUMAN AI

Two short texts on the same topic: "Why our school library should stay open after class." One was written by a Class 8 student; one was generated by an AI text tool. Read like an auditor — then give your verdict.

**THE HUMAN WAY**

Essay A: "Our library closes at 4:30, but last Tuesday I stayed back for a maths test and saw three juniors asking the guard to let them in to return books. Opening till 6 on two days would fix exactly that. It costs almost nothing — the guard is already there."

**THE MACHINE WAY**

Essay B: "Libraries are the heart of every school. Keeping the library open after class offers numerous benefits: it fosters a love of reading, provides a quiet study environment, and ensures equitable access to resources for all students. In today's fast-paced world, such spaces matter more than ever."

My verdict — A or B — and the ONE feature that gave it away (a real event? a vague phrase? whose voice?):

Swap test: rewrite one sentence of Essay B so a reader can tell a human wrote it. What did you add?

Two "tell-tale AI phrases" I will watch for in text I read this week:

## 4.3 The Responsible-Use Court: decisions with consequences

Technology choices are people choices. A school, a company or a government decides **what** a machine may do, **to whom**, and **who answers when it fails**. These decisions deserve the same seriousness as any other — evidence, arguments, and a verdict someone signs. So: court is now in session.

**COURT PREP — MY FIRST CASE NOMINATION**

Before the session opens: which real AI decision near you would you bring to this court first, and who should sit in the jury?

### T8-13 Responsible-Use Court

CLASS · 25 MIN

**The case:** Sunapur school installed a "focus camera" in Class 8-B that watches faces and flags students who "look distracted" to the teacher every 10 minutes. The vendor says it raises results. Two students say it feels like being suspected all day. One parent says her son's medication makes him look drowsy — the machine flags him constantly.

Assign roles: one **Judge**, two **Defence** lawyers (for the camera), two **Prosecution** lawyers (against), two **Witnesses** (the parent; a Class 8-B student), and the rest are the **Jury**. Each side gets 3 minutes; witnesses 1 minute each. Then everyone fills the verdict sheet:

| Court question | The defence's best point | The prosecution's best point |
|---|---|---|
| Does the machine measure what "focus" really is? | | |
| Who is harmed if it is wrong — and how? | | |
| Was consent taken from the people watched? | | |
| Is there a less invasive way to the same goal? | | |

**THE JURY'S VERDICT (pick one and give the reason):** keep the camera · keep with changes (name them) · remove it.

WORD POWER · 16

Definition: **responsible use**

Choosing and using AI tools so people are helped, respected and safe.

*Read it three times today — it sticks!*

## 4.4 My AI-Use Pledge

Court cases end with judgements; engineer stories end with promises. A pledge is not decoration — it is the checklist you write **before** the difficult moment arrives, when a chatbot offers to write your essay or a group-chat spreads an exciting rumination. Copy your class's best ideas into the box, sign it, and keep the page where you will meet it again at the end of the book.

## T8-14 My AI-Use Pledge

ALONE · 10 MIN

Complete each clause honestly — these are YOUR rules, not the teacher's:

**I, the undersigned engineer, pledge that…**

1. Before I believe or forward any AI-written or AI-edited content, I will

2. When a text generator helps me with schoolwork, I will

3. When I am tempted to feed personal information — mine or anyone's — into an AI tool, I will

4. When an AI tool makes a decision about a person, I will ask the two court questions:

5. If a machine's output hurts someone, I will hold responsible the … (circle one: machine / its user / its maker / whoever had control) — **because**

**Signature:** _____ **Date:** _____

### Self-check

shade one: first circle = not yet · middle = almost · last = got it

- ☐ I can explain in two sentences how a text generator writes — and why it can invent ○○○
- ☐ I can run the three-move routine — trace, test, smell — on a suspicious claim ○○○
- ☐ I can argue both sides of a responsible-use case and sign a reasoned verdict ○○○
- ☐ I can name what my own pledge demands of me before the difficult moment arrives ○○○

**THINK & INK**

*A chatbot once confidently told me something that turned out to be wrong (or would have, if I had believed it). Now I understand why that happened: …*

**HOME LINK** *take this page home — no screen needed*

**AT HOME** **Family Fact Patrol** HOME · 15 MIN

With an adult, find one forwarded message or AI-looking summary in the family phone (do not open links). Run the routine together: **trace** — where did it really come from? **test** — do two trusted sources agree? **smell** — does it push an emotion before evidence?

**The claim we examined:**

**What the three moves found:**

**What we will do differently before forwarding next time:**

**CASE NOTES — WHAT CHANGED IN MY THINKING?**

Before this chapter I believed a confident, fluent answer was usually a correct answer. Now I know fluency and truth are built differently, because …

**CHAPTER 4 CLUES — POCKET THESE**

- A text generator **predicts likely words** — likely is not the same as true.
- **Hallucinations** are built from the same fluency as facts — verify with trace, test, smell.
- Responsible use is decided by **people** — in courtrooms, staffrooms and pledges, with reasons.
- Responsibility follows **control**: machines control nothing, so people answer — makers, users, deployers.

**The clue I would tell my family tonight:**

**MY COURT RECORD — THE VERDICT I SIGNED**

Case: the focus camera. I argued for the side of … _____ · my verdict: _____ · one argument that changed my mind mid-debate:

## Chapter 4 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

**1.** Why does a text generator hallucinate — what exactly is it doing instead of checking facts?

**2.** A forwarded message says “92% of parents demanded a focus camera”. Which of the three moves do you run first, and what do you look for?

**3.** The court’s hardest question: the camera’s **maker**, the school that **installed** it, and the teacher who **uses** it — who answers when it wrongly flags a student? Why?

**MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT ETHICS AND RESPONSIBILITY**

# 5. Capstone: The AI Project Proposal

**GUIDING QUESTION**

*Can you plan an AI that people can trust — and defend the plan to a live audience?*

The final case brings every clue together. Your team will write a complete AI project proposal on paper — sharp problem, audited data plan, chosen tool, honest test, fairness check and known limits — exactly the document a professional team brings to its first review. Then your classmates will pressure-test it in a peer review gallery, and you will sharpen your budget choices with a real optimisation problem. This is not an exam; it is the moment the investigator becomes the engineer.

**MISSIONS** T8-15 – T8-17
**OUTCOMES** 8.U1 · 8.L1 · 8.L2 · 8.D1 · 8.D2 · 8.W1 · 8.W2 · 8.R1 · 8.R2 · 8.T1 **STRANDS** U D L W R

*Links: all five strands · English — persuasive writing · Maths — budgeting & optimisation*

**In this chapter**


5.1 The proposal: your plan on paper ..... 59
5.2 The Peer Review Gallery ..... 61
5.3 Optimise It: the best you can afford ..... 63


**TERMS YOU'LL BUILD WITH**

peer review · trade-off · optimise · stakeholder

**WARM-UP · BEFORE YOU READ** ALONE · 2 MIN

You have one page and one minute to convince your class to build your AI. What is the FIRST thing you would say — the problem, the data, or the dream? Write it, then check if your answer survives Task T8-15.

**ENGINEER'S TOOLKIT — YOU WILL NEED:** your whole notebook · pencil AND eraser · the fairness checklist from Chapter 2 · courage to share

## 5.1 The proposal: your plan on paper

Every serious AI project begins as a document nobody coded: a proposal. Its job is to answer six questions **before** anyone builds — What problem, exactly? What data, from whom, with what consent? Which tool, and why that one? How will we test it honestly? Who could it hurt, and what did we change for them? What can it NOT do? A proposal that answers all six can still fail — but a project without one fails **surprised**, which is worse. Your proposal lives in Tasks T8-15 over the next pages. Choose a real, local, small problem: your classroom, your street, your school's morning chaos. Small and real beats grand and imaginary every time.

### T8-15 · Part 1 AI Project Proposal — the plan

GROUP · 3 periods MIN

Part 1 of 2. Work as a project team of three or four. Choose your problem, then scope it with the discipline of Chapter 1.

**PROPOSAL · SECTION A — THE SHARP PROBLEM**

**Our problem statement (passes the 4W test):**

**The stakeholders — everyone this machine would touch, helped or harmed (name at least three groups):**

**Why existing solutions are not enough:**

**SECTION B — THE TOOL**

**Which no-code tool type fits — classifier, chatbot or predictor — and why that one:**

Its input, its output, and one hard limit we accept:

**T8-15 · Part 2** **AI Project Proposal — data, test & fairness** GROUP · 3 periods MIN

Part 2 of 2. This is where the Plant Doctor failed. Do not skip a box.

## SECTION C — THE DATA PLAN

| What we collect | From whom (all the groups!) | How we get consent | How we protect privacy |
|---|---|---|---|
| | | | |
| | | | |

Our guest-list check (from Chapter 2): which groups must be INSIDE the data for this to be fair — and how many from each?

## SECTION D — THE HONEST TEST

Our hidden test set: how many cases, and why that number:

The accuracy we would accept before shipping (and the accuracy that would send us back to the drawing board):

## SECTION E — FAIRNESS & LIMITS

The group most likely to be failed by version 1 (be specific):

**What we changed in the plan because of that group:**

**What our machine will NEVER do (write it down before someone assumes otherwise):**

## 5.2 The Peer Review Gallery

Real proposals are never shipped on the engineer's word alone — they face review. In professional teams, reviewers hunt for the **weakest box**, not the prettiest sentence, and the proposal gets better because of it. Today your classmates are your reviewers: every plan is pinned up (or read aloud), every team walks the gallery with a clipboard, and every review ends with one strength, one gap, and one question the engineers cannot dodge.

**GALLERY MARSHAL — HOUSE RULES:** **review the PLAN, never the people · one strength before any gap · every gap must come with the question that would close it · engineers may only reply with evidence, not feelings.**

**THE WEAKEST BOX IN AI HISTORY — a 30-second recall:** The Plant Doctor proposal would have scored well on THREE of the six boxes. Which box would you have graded Beginning — and what ONE line would have saved it?

**MY GALLERY CLIPBOARD — plan the walk before you walk it**

The proposals I will review (numbers): _____ _____ _____ The question I am proudest to ask:

**Reviewer's oath:** "I will find the weakest box, say what is strong first, and leave every plan better than I found it."

## T8-16 Peer Review Gallery

CLASS · 20 MIN

Walk the gallery. For each proposal except your own, fill one review card. Be honest and kind — you are reviewing the **plan**, never the people:

**REVIEW CARD — FOR PROPOSAL № _____**

**The strength I would praise first:**

**The weakest box right now (problem / data / test / fairness):**

**The one question the engineers cannot dodge:**

**Rubric shade for this proposal** — Beginning / Growing / Secure / Shining **(shade one)**
○○○○

**Now the reverse mirror:** the sharpest question a reviewer asked about YOUR plan, and what version 2 will do about it:

**CARD 2 OF 3 — THE ONE-MINUTE VERSION (use for a third proposal)**

Proposal № _____ · weakest box: _____ · rubric shade: ○○○○

**My closing question for the engineers:**

**THE FOUR LEVELS — HOW THIS BOOK GRADES REASONING**

| Level | What it looks like in a proposal |
|---|---|
| Beginning | The problem is a slogan; the data is "we will find some"; fairness is one word. |
| Growing | Real problem and tool chosen, but the test is vague or the fairness check is a promise, not a plan. |
| Secure | All six boxes are answered with evidence — and the limits are written before anyone asked. |
| Shining | Secure, PLUS the team found a trade-off nobody noticed and made a defensible choice anyway. |

*Notice: a flawed answer with honest reasoning can outscore a polished answer with no reasons. This rubric rewards exactly what engineers get paid for.*

## 5.3 Optimise It: the best you can afford

One last thinking-tool before graduation. Real engineers never have enough budget for everything — they **optimise**: they break the choice into parts (cost, benefit, fairness) and find the best combination under the limit. This is computational thinking in its purest form: no AI at all, just you, a table and a constraint.

**THE OPTIMISER'S HABIT**

Break the choice into parts (this item costs this, buys that), compute honestly, then defend what you leave out. The defence matters as much as the choice — a budget nobody can argue with is a budget nobody checked.

**BUDGET ARITHMETIC WARM-UP — BEFORE YOU SHOP**

Add up these three starter combinations. Which ones fit inside ₹500 — and which item appears in BOTH affordable pairs?

| Combination | Total cost | Fits ₹500? |
|---|---|---|
| A + B (trip + test cards) | | |
| C + D (two features + lighting) | | |
| B + C + D (tests + features + lighting) | | |

## T8-17 Optimise It

ALONE · 10 MIN

Your club has **₹500** to improve the leaf classifier before its second test. Choose the combination that gets the most trustworthy machine. You may buy each item at most once. Warm-up totals: A+B = ₹300 · C+D = ₹400 · B+C+D = ₹500 — all three fit, but none fixes everything. Now choose:

| Upgrade item | Cost | What it buys you |
|---|---|---|
| A · Second field trip for 20 more leaves, half from the river farm | ₹200 | fixes the missing-group problem in training |
| B · Print 10 extra hidden test cards | ₹100 | bigger test set — a steadier accuracy number |
| C · Measure both leaf features (edge AND length) for all cards | ₹150 | two features instead of one — fewer mix-ups |
| D · Photograph everything in shade AND sun | ₹250 | kills the lighting weakness the Plant Doctor had |

**My combination (letters + total cost): _____ The accuracy problems this combination fixes:**

**The weakness I knowingly leave unfixed — and my defence for leaving it:**

**Could any combination fix everything at once? What does that tell you about real projects?**

**MYTH BUSTER**

**MYTH** *"The model knows why."*

**TRUTH** Ask the leaf classifier **why** it called a leaf type B, and it has no answer — only distances. Ask the text generator why it wrote a sentence, and it will **invent** a plausible reason, the same way it writes everything else. The "why" lives in three places, and none of them is the model: in the **data** (what patterns existed), in the **design choices** (which features, which examples), and in the **people** (who decided what it was for). That is why every chapter of this book made you write your reasons beside your answers — the reasoning was never the machine's job. It was, and is, yours.

## Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can write a proposal that answers all six engineer questions with evidence ○ ○ ○
- [ ] I can review a classmate's plan with one strength, one gap and one unavoidable question ○ ○ ○
- [ ] I can grade a proposal — mine or a classmate's — with the four levels and say why ○ ○ ○
- [ ] I can choose the best combination under a budget and defend the trade-off I accept ○ ○ ○

**THINK & INK**

*After this year, my honest definition of "trustworthy AI" is … — and one thing I now check that I never checked before is …*

**CASE NOTES — WHAT CHANGED IN MY THINKING?**

Before this year I believed AI was mainly about clever machines. After building (on paper) my own, I believe it is mainly about …

**CHAPTER 5 CLUES — POCKET THESE**

- A proposal answers six questions **before** building: problem, data, tool, test, fairness, limits.
- **Peer review** hunts the weakest box — the plan that survives honest reviewers earns its trust.
- **Optimising** means choosing the best combination under limits — and defending what you leave out.
- The model never knows why — the reasons live in the data, the design and the people.

**The clue I would tell my family tonight:**

## Chapter 5 Checkpoint

one line each — no peeking back!

**ALONE · 3 MIN**

**1.** Name the six sections of your proposal, in order — no peeking back.

**2.** A reviewer shades “Growing” on a proposal whose boxes are all filled. What is most likely missing?

**3.** In T8-17, why is “fix everything” impossible — and what does a real team do about that?

**MY CASE JOURNAL — THE LAST CLUE OF CASE 3, FOR THE ROAD TO CLASS 9**

# My AI words

*The sixteen words on this case — each one met through an experience, never before.*

**1 problem statement**
One precise sentence saying exactly what problem you solve, for whom, where.
USE IT IN MY OWN SENTENCE:

**2 AI project cycle**
The loop: scope the problem, collect data, build and test, reflect and improve.
USE IT IN MY OWN SENTENCE:

**3 dataset**
The collection of examples a machine learns from, or is tested on.
USE IT IN MY OWN SENTENCE:

**4 sampling**
Choosing part of a group to study, hoping it fairly represents the whole.
USE IT IN MY OWN SENTENCE:

**5 representation**
How closely a dataset's people match the real people the system will serve.
USE IT IN MY OWN SENTENCE:

**6 fairness**
A system works equally well for every group of people it affects.
USE IT IN MY OWN SENTENCE:

**7 consent**
Permission given freely by a person who understands what will be collected and why.
USE IT IN MY OWN SENTENCE:

**8 privacy**
Keeping people's personal details safe, used only for the promised purpose.
USE IT IN MY OWN SENTENCE:

**9 nearest neighbour**
Classifying a new example by its most similar labelled examples.
USE IT IN MY OWN SENTENCE:

**10 accuracy**
The share of test examples a machine gets right, written as a percentage.
USE IT IN MY OWN SENTENCE:

**11 no-code tool**
A ready-made AI tool you train or use without writing any program.
USE IT IN MY OWN SENTENCE:

**12 classifier**
A machine that sorts examples into named groups, like ripe or unripe.
USE IT IN MY OWN SENTENCE:

**13 text generator**
A machine that writes by predicting the most likely next words.
USE IT IN MY OWN SENTENCE:

**14 hallucination**
When a text generator states something false as if it were fact.
USE IT IN MY OWN SENTENCE:

**15 misinformation**
False content that spreads — by mistake or on purpose.
USE IT IN MY OWN SENTENCE:

**16 responsible use**
Choosing and using AI tools so people are helped, respected and safe.
USE IT IN MY OWN SENTENCE:

**ENGINEER'S OATH**

Machines **compare, predict, continue** — people **scope the problem, audit the guest list, run the honest test and sign the verdict**. When a machine speaks, I will ask my five questions: **How do I know? What is missing? Who made this? Who is missing? What is the trick here?**

### ANSWER HINTS — FOR CHECKING, NEVER FOR COPYING

**T8-04:** Ward-3 adults are 120 ÷ 200 = **60%** of the dataset against an 18% town share — overweighted more than three times; elderly are **missing** (0%); Kannada-only speakers are 2% against 22% — nearly missing. · **T8-05:** the hand-raised sample is biased before any number is written — quick hands, front rows and confident friends are not a random draw. · **Name the stage drill:** P, D, M, T, I, P. · **Sharpen or spoil:** S, V, S, V — the vague ones are missing all four Ws at once. · **Skewed sample drill:** S, S, F, S, S — the lunch-bag plan only hears from packers and repliers.

### MORE HINTS — FOR THE STICKY MISSIONS

**T8-08:** Bird 1 and Bird 2 both fall among the teal dots — POND. Bird 3's nearest dot is the amber one at about (8.5, 6.2) — FOREST — but write down which **feature** actually did the deciding, because the vibes disagreed. · **T8-09:** timings → chatbot; rice demand and homework-help flags → predictor; cracked panels → image classifier. · **Three-move drill:** N, S, T, N, S. · **Accuracy drill:** 85%, 94%, 90% — then ask what the sunny-day-only test set hides. · **T8-17:** every four-item combination costs ₹700, so **no** plan fixes everything — that is the lesson. Strong plans include A (missing groups) and then defend what they left out.

### MY HINT LOG — WHICH MISSIONS NEEDED A PEEK?

Hints are training data for you. Tick the mission code each time a hint unstuck you — a mission that needed three hints deserves one more retry next week, not three more hints:

- ☐ T8-__ needed a hint
- ☐ T8-__ needed a hint
- ☐ T8-__ needed a hint
- ☐ T8-__ needed a hint

### A POSTCARD TO THE CLASS 9 ANALYST — WHO WILL BE YOU

The next case asks the question every engineer must face sooner or later: **"Why does AI work — and when does it fail?"** Class 9 goes under the hood: the maths of averages, probability and best-fit lines that make learning possible, and the graph tricks that make it look easier than it is. Leave three clues for your future self — the trade-off that surprised you most, the plan of yours that failed hardest, and one question you dare Class 9 to answer:

# My progress

*Shade honestly — engineers never fake evidence.*

### MY OUTCOME TRACKER — TICK WHEN YOU TRULY CAN DO IT

- ☐ 8.U1 — define AI and machine learning in my own words; place chatbots, classifiers and recommenders on the who-learns-what map
- ☐ 8.L1 — walk a local problem through the full AI project cycle, on paper
- ☐ 8.L2 — hand-run a nearest-example classifier, compute its accuracy, and suggest an improvement
- ☐ 8.D1 — audit a small dataset for missing groups and compute percentages per group
- ☐ 8.D2 — explain why collecting data needs consent and privacy care
- ☐ 8.W1 — match no-code tool types to problems, stating inputs, outputs and limits
- ☐ 8.W2 — explain why a text generator can sound right yet be wrong — and verify its claims
- ☐ 8.R1 — apply a fairness checklist and propose fixes for an unfair dataset
- ☐ 8.R2 — make and defend a responsible-use decision about privacy, misinformation or impact
- ☐ 8.T1 — break a multi-variable problem into parts and find the best choice under limits

### MY CASE FILES — SHADE A STAR PER CHAPTER FINISHED

*1 Project Cycle · 2 Data & Fairness · 3 How Models Learn · 4 Responsible AI · 5 Proposal*



PRATIMAI · AI HANDOUTS · CLASS 8

## My AI Engineer's Certificate

*This certifies that project lead*

_____

ENGINEER NAME

has completed the case file **"Build the AI Cycle"** — Level 3 · BUILD THE CYCLE — planned and peer-reviewed a full AI project proposal, and promised to keep asking: **How do I know? · What is missing? · Who made this? · Who is missing? · What is the trick here?**

| _____ | _____ | _____ |
|---|---|---|
| DATE | ENGINEER'S SIGNATURE | GUIDE'S SIGNATURE |

## My series so far — three cases, one engineer

**CLASS 6 · NOTICE**
I learned to **spot** learning machines and to ask: how do I know?

**CLASS 7 · SORT & PREDICT**
I learned to **train and test** machines — and how examples fool them.

**CLASS 8 · BUILD THE CYCLE**
I learned to **plan, audit and defend** an AI before it is built.

**The habit I am proudest of now:**

## Where I used my five questions — the tally of a finished case

| My question | Where I asked it for real (at home, in class, online) |
|---|---|
| How do I know? | |
| What is missing? | |
| Who made this? | |
| Who is missing? | |
| What is the trick here? | |

*Five rows is the whole method. If a row is still empty, the next case (Class 9) is where you will catch up.*

**THIS CASE FILE BELONGS TO THE EVIDENCE**

Engineer: _____ Class & division: _____ School year: _____

*Hand this book to next year's Class 8 engineers with your tallies inside — the best hand-me-down is an honest one.*

**THE DOOR TO CLASS 9**

Next year the case goes **under the hood**: the averages, probability and best-fit lines that make machine learning possible — and the graph tricks that make it look easier than it is. Keep this book; its checkpoints are your revision notes. And whatever machine you meet next, keep asking your five questions:

**How do I know? · What is missing? · Who made this? · Who is missing? · What is the trick here?**

Ready for the next case? Class 9 asks: **"Why does AI work — and when does it fail?"** — bring your blueprint.